

Rohit Kumar Birakayala

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Experience

Data Scientist Intern

Gracenote, Nielsen

Bengaluru, India
Jan 2026 – Present

- Improved version-match precision from 48% to 87% by engineering a version-aware media matching pipeline using Claude Haiku/ Sonnet hosted on Amazon EC2 with semantic label extraction
- Performed DSPY prompt optimization with COPRO, MIPRO & InferRules optimizers researched via the Context7 MCP server, evaluated on curated GIST sampled datasets
- Curated indexed parquet datasets from 10k+ production records and ground-truth Program IDs, enabling scalable retrieval across 10k+ incoming mappables and avoiding infeasible $O(n^2)$ search
- Engineered an RCA-driven feedback pipeline for multilingual LLM translations using a 2-stage actionable record filtering system, clustering of recurring rule violations, updating an evolving knowledge base reducing editor violations by 25+%

Software Engineering Intern

UBS – GOTO Technology

Pune, India
May 2025 – Jul 2025

- Automated classification workflows in ServiceNow using Predictive Intelligence, achieving over 80% precision and recall while applying boosting techniques to mitigate skewed data and class imbalance.
- Optimised prediction performance by identifying the 4 highest impact input features through targeted data analysis, improving automation by 20%
- Integrated JavaScript modules with ServiceNow UI to retrieve and present predictive insights

Machine Learning Intern

Talent Recruit

Bengaluru, India
May 2024 – Jul 2024

- Developed a RAG pipeline for matching top-k resumes with job descriptions using semantic similarity, incorporating MiniLM and BERT embeddings indexed in ChromaDB for an efficient vector search.
- Built a SpanCat-based parser for skill extraction and boosted system accuracy by 15%+ through generating 2000+ hard negative samples using similarity thresholds and manual scoring.

Projects

Group Expense Splitter App

[GitHub](#) | [WebApp](#)

- Designed a Flutter app with Firebase backend to manage group trip expenses, member balances, and settlements
- Implemented graph-based settlement optimization to eliminate transitive paybacks that minimize repayment transactions

4 Phase Sickle Cell Prediction

- Researched and built a 4-stage medical imaging pipeline combining image segmentation, morphological extraction of 30 key features, PCA based dimensionality reduction, GMM clustering followed by image level classification.
- Achieved 98%+ precision and recall, outperforming benchmark architectures including DenseNet and InceptionV3.

Technical Skills

Languages: Python, Java, C++, JavaScript, SQL, Dart, TypeScript

AI/ML: LLMs, NLP, RAG pipelines, Prompt Optimization, Semantic Similarity, Computer Vision, MultiAgent systems

Frameworks: TensorFlow, PyTorch, Keras, Streamlit, Flutter, Git, ServiceNow, DSPY

Data & Infra: ChromaDB, MongoDB, PostgreSQL, Parquet, Vector Search, Indexed Retrieval, Amazon EC2

Tools & Platforms: Git, ServiceNOW, Jira Webhooks, CrewAI, Context7 MCP Server

Certifications: 3 specialization Machine Learning courses by Stanford University & DeepLearning.AI

Education

Vellore Institute of Technology

B.Tech in Computer Science with specialization in Data Science

Sept 2022 – Jun 2026
GPA: 9.5

Geetanjali Olympiad School 12th CBSE - 93%

Jun 2020 – Jun 2022

New Horizon Public School 10th ICSE - 98%

Apr 2010 – Jun 2020

Positions of Responsibility

Vice Chairperson, Juvenile Care NGO, VIT

Jan 2025 – Jan 2026

- Spearheaded planning and execution of 10+ events in collaboration with external NGOs and VIT's Welfare Office, serving 500+ attendees and promoting community engagement.